Week 6. By Deng Yufan.

Prob 1. Prove if $P = NP$. then $NP = coNP$.

Since $P \subseteq coNP$, as for a P prob., the solver itself can be a verifier, we have $NP \subseteq coNP$.

For a $coNP$ prob.,

For a P problem L, we have $\bar{L} \in P$ since we can swap the solver's accept and reject.

So if $P = NP$. for a NP problem L. we have $\bar{L} \in NP$, thus $L \in coNP$;

for a $coNP$ problem L, we have $\bar{L} \in NP$, thus $L \in NP$.

Prob 2. For 2-SAT prob., draw a directed graph of $2n$ vertices. Prove the instance is unsatisfiable iff x_i and $\neg x_i$ in same SCC. Show 2-SAT $\in P$.

If: Obviously if every clause can be satisfied, then $x_i \to \neg x_i$, $\neg x_i \to x_i$,

thus $x_i \leftrightarrow \neg x_i$, contradiction.

Only if: If all x_i and $\neg x_i$ are not in the same SCC (Strongly Connected Component),

then for i from 1 to n. if x_i and $\neg x_i$ both unpicked. pick x_i

and all x_j (or $\neg x_j$) that in the same SCC as x_i.

If x_i is picked, set x_i to 1. Otherwise set to 0. Obviously it is a valid solution for the instance.

To show 2-SAT $\in P$, we give an algo to solve 2-SAT in P.

1°. Build the graph just mentioned. ~~$O(m)$~~. $O(n+m)$

2° Use Tarjan algorithm to find all local maximal SCC. ~~$O(m)$~~. $O(n+m)$.

3° Judge if x_i and $\neg x_i$ in the same SCC. $O(n)$.

4°. If $\exists i$. x_i and $\neg x_i$ in the same SCC, reject.

Other-wise construct a solution by the approach in "Only if" part. then accept. $O(n)$.

So we solve the problem in $O(n+m)$.

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Prob 3: Lehmer's Thm: $n \in N$ is prime iff $\exists a.\ a^{n-1} \equiv 1 \pmod n$.
$\forall q | n-1.\ a^{(n-1)/q} \not\equiv 1 \pmod n$.

Show that PRIME $\in$ NP $\cap$ coNP.

To show PRIME $\in$ coNP. just consider verifier that given n, x, y, and check $x \neq 1.\ y \neq 1.\ n = xy$. This can be done in $O(|n| \log |n|)$. since $x, y < n$.

To show PRIME $\in$ NP, consider verifier that given $n, a, [q_1, q_2, \cdots, q_k]$, satisfy $a, q_1, \cdots, q_k < n$, $2 \le k \le \log n = |n|$. and

To show PRIME $\in$ NP. define witness for n is $[n] := a, k, [q_1], [q_2], \cdots, [q_k]$ which s.t. $0 \le a < n$. $a^{n-1} \equiv 1 \pmod n$. $\forall 1 \le i \le k$. $a^{(n-1)/q_i} \not\equiv 1 \pmod n$.

and $[q_i]$ is prime factoring for $n-1$. If $n-1 = 1$, $k = 0$.

Thus verifier only need to verify the condition above.

For $a^{n-1} \equiv 1 \pmod n$. use quick power can done in $O(|n|^2 \log |n|)$.

For $[q_i]$ is prime factoring. since $q_i \ge 2$, $k < \log n$, simply multiply q_i takes $O(|n|^2 \log |n|)$.

Lastly we need to check the witness for $q_1, \cdots, q_k$.

Then the time complexity is $T(|n|) = O(|n|^2 \log |n|) + \sum_{i=1}^{k} T(|q_i|)$.

Since $O(|n|^2 \log |n|)$ is convex function. $\sum_{i=1}^{k} T(|q_i|)$ reaches maximum when $k = 1$. $q_1 = n-1$ is prime. Thus

$T(n) \le O(|n|^2 \log |n|) +$

Easy to see $k = 0$ only when $n = 2$. $k = 1$ only when $n = 3$. So we can suppose $k \ge 2$.

Since $O(|n|^2 \log |n|)$ is convex. we have

$$\begin{aligned} T(|n|) &\le O(|n|^2 \log |n|) + T(|q_1|) + T(|q_2|) \\ &\le O(|n|^2 \log |n|) + T(|2|) + T(|\tfrac{n-1}{2}|) \\ &\le O(|n|^2 \log |n|) + T(1) + T(|n|-1). \\ &\le \sum_{i=1}^{|n|} O(i^2 \log i). \\ &= O(|n|^3 \log |n|). \quad \text{By calculus.} \end{aligned}$$

Thus we can verify $[n]$ in $O(|n|^3 \log |n|)$.

Thus the length of $[n]$ won't beyond $O(|n|^3 \log |n|)$. ✓

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